

# 1. Introduction

This document is a guided tour of `mos-lsp`, the Mosaic language server. It is also an ordinary document: the headings are numbered and the prose cross-links to them. Section 2 lists what the server can do, section 3 covers installation, and the closing remarks sit in section 4 on page 2.

Each cross-reference above prints as a number in the PDF, but in the `.mos` source it is a live `@label` reference. That duality is the whole point: readers of the PDF get clean section numbers, while an editor wired to `mos-lsp` can navigate and rename the same references in the source.

## 2. Language server features

`mos-lsp` ships three capabilities today:

- Diagnostics surface compiler errors as you edit.
- Go-to-definition jumps from a reference to the heading that declares its label.
- Rename rewrites a label's declaration and every reference together.

Citations resolve too: the nod to the classic typesetting manual [1] and the bibliography-style reference [2] print as numbered markers.

### 2.1. Go-to-definition

Go-to-definition jumps from a reference to the heading that declares its label. This section demonstrates it on itself: it holds a live reference to its own heading, section 2.1, which the PDF prints as the number you see. In the source that token is `@gotodef`; put the cursor on it and run `go-to-definition` to land on this heading's `<gotodef>` label.

### 2.2. Rename

Rename rewrites a label's declaration and every reference to it in one edit. This heading declares the `<rename>` label, and the cross-link here, section 2.2, is a reference to it. Rename either the declaration or this reference, and both change together, in view, without scrolling away from this section.

### 2.3. Diagnostics

To watch diagnostics appear, introduce an error in the source: point a reference at a label that does not exist, or cite an unknown bibliography key. `mos-lsp` reports the problem inline; undo to clear it.

## 3. Setup

Install the server so it lands on `PATH`: run `cargo mosi1s` from the workspace, then

open this project in your editor. Section 1 has the overview and section 4 has the final check.

## **4. Conclusion**

If go-to-definition reached this heading from a reference, rename swept a label cleanly, and the page printed for section 4 matches where you are reading, the language server is wired correctly.